

## Smart Security Safe System



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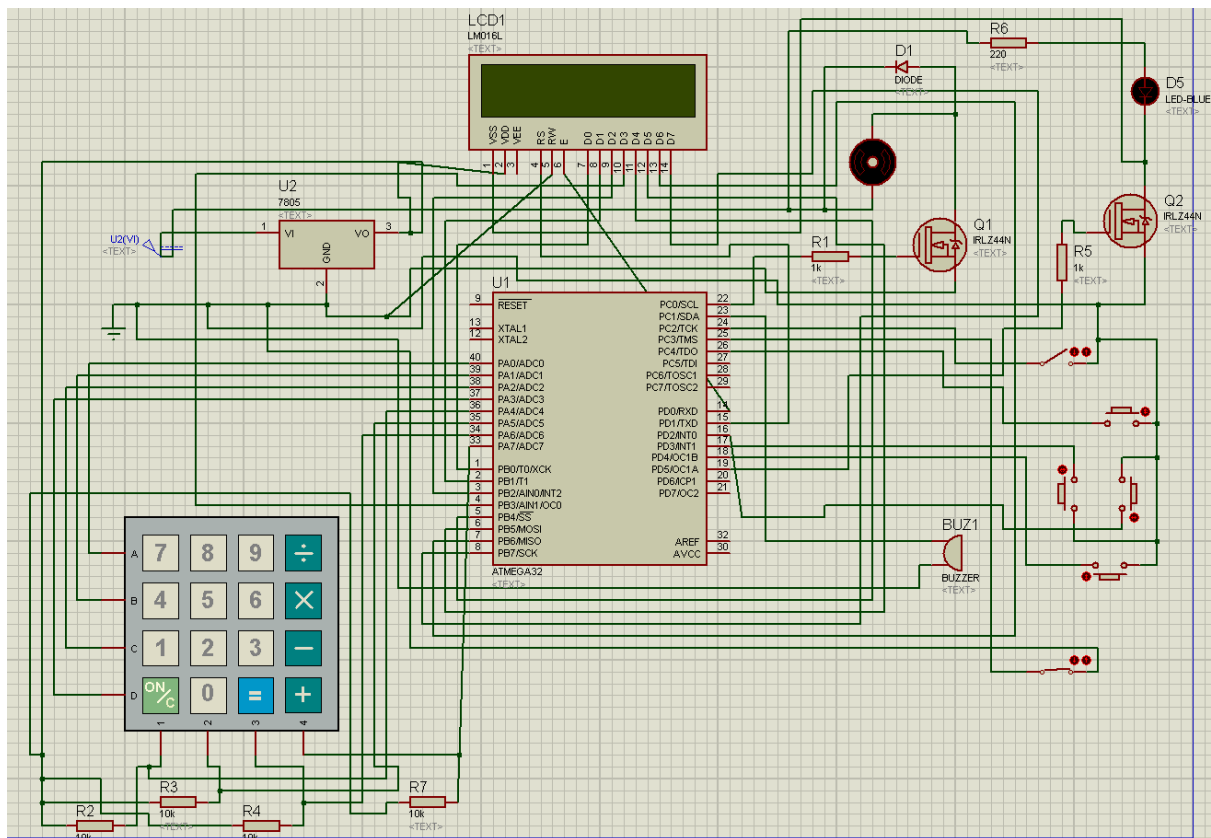
## Project Overview

This project is a smart electronic safe system designed to provide multiple layers of security and a user-friendly control interface. The safe can be operated using either a password or a physical key for authentication and security management.

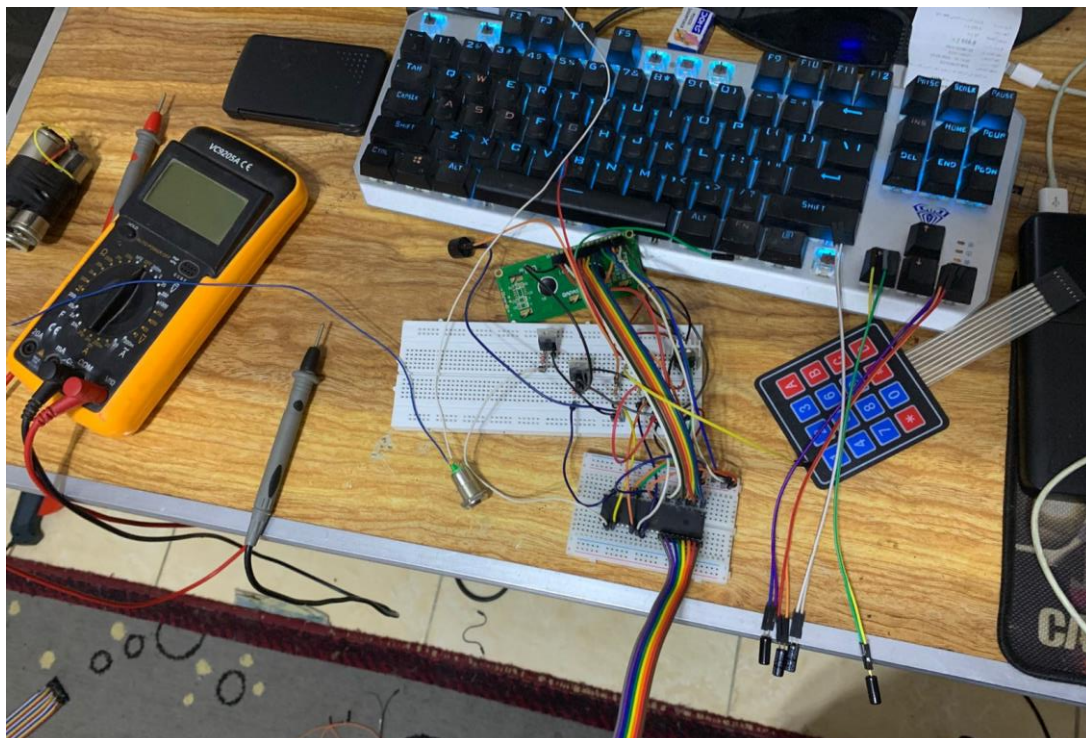
## Main Features

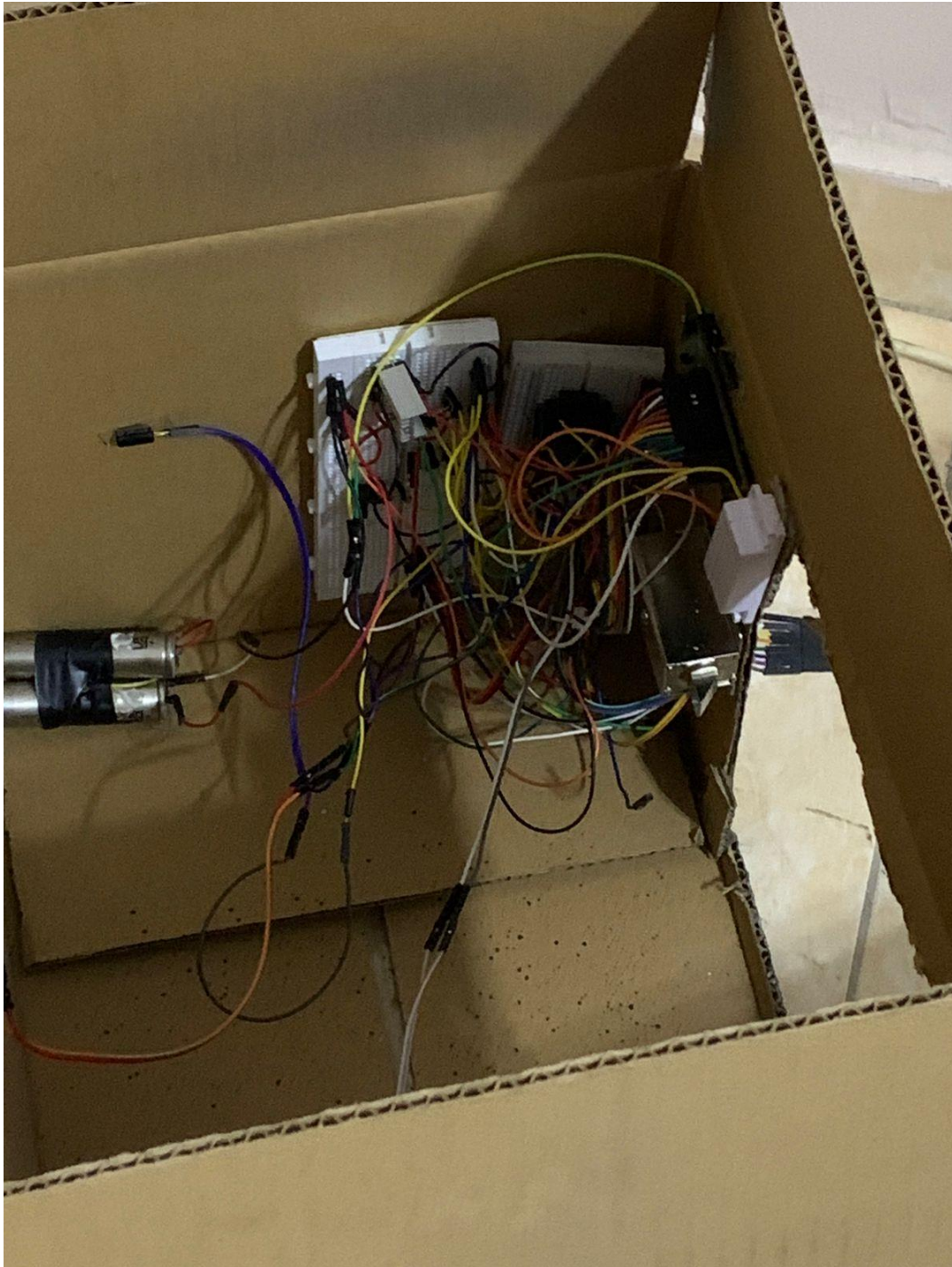
- Open the safe using a password or a physical key.
- Change the password using either password authentication or the physical key.
- Enable or disable transportation and anti-theft protection mode using either the password or the physical key.
- Navigate easily through a menu-based LCD interface.
- The password is securely stored in EEPROM memory to preserve data even after power loss.

## System Overview



Proteus Design





**Hardware Implementation**

## **LCD and User Interface**

**The system uses an LCD display with several smart power-saving and usability features:**

- If the user stays on the main menu without pressing any keypad button for three menu cycles, the LCD backlight turns off automatically to save power.
- Pressing any keypad button turns the LCD backlight on again
- If the user stays inside any option (such as opening the safe, changing the password, or protection settings) for 20 seconds without interaction, the system automatically returns to the main menu.
- Password characters are hidden as \* symbols while typing.
- Pressing \* toggles password visibility between hidden stars and visible numbers.

- Pressing # deletes the last entered digit from the password input.
- Pressing B at any time returns the user to the main menu.
- The system provides interactive sound feedback when buttons are pressed or when an incorrect password is entered.

## **Security System**

**The safe includes multiple protection mechanisms to prevent unauthorized access or theft:**

- Four SW-520D vibration sensors are installed on the four sides of the safe to detect movement, carrying attempts, or breaking attempts.
- A magnetic switch is installed on the safe door to detect unauthorized opening.
- If the safe door is opened without using the correct password or key, the alarm is activated immediately.
- Transportation and anti-theft protection mode can be enabled or disabled using either the password or the physical key.
- The four vibration sensors are distributed on all four sides of the safe for full coverage detection.

## **Alarm System**

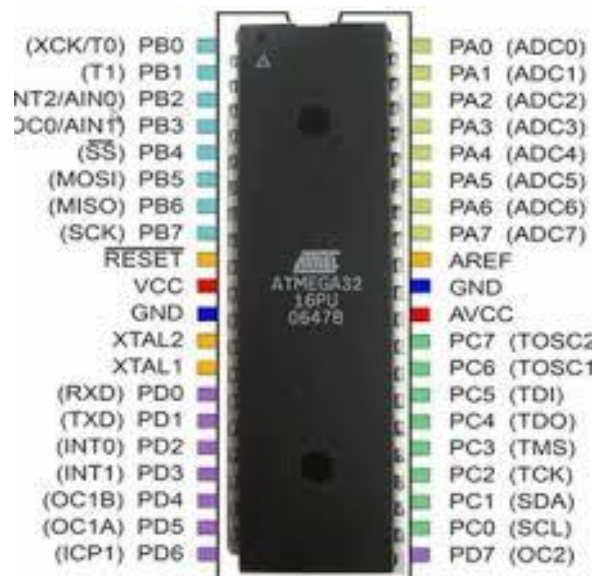
**The alarm is triggered in the following situations:**

- Entering the wrong password three consecutive times.
- Detecting vibration or movement while protection mode is enabled.
- Detecting unauthorized door opening using the magnetic switch.

**Once activated, the alarm can only be turned off using the physical key for maximum security.**

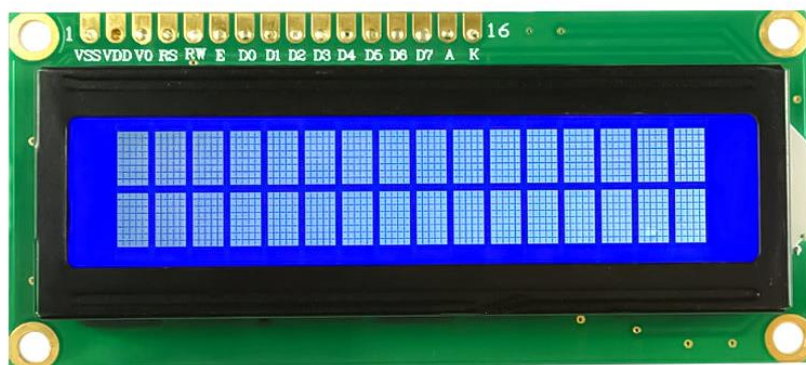
## Hardware Components

1:



**ATmega32 Microcontroller**

2:



**16x2 LCD Display**

3:



**4x4 Keypad**

4:



**4x SW-520D Vibration Sensors**

5:



**Magnetic Door Switch**

6:



**Buzzer**

7:



**12V DC Lock**

8:



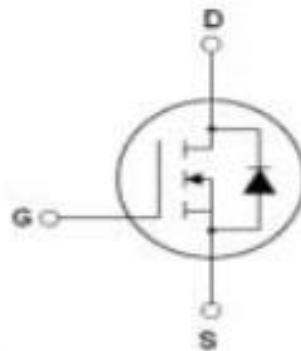
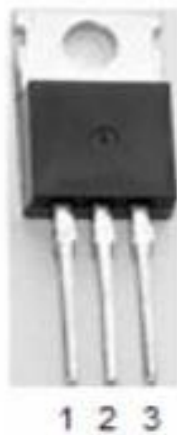
**12V Battery**

9:



**Key Switch**

10:



pin 1, Gate  
2, Drain  
3, Source

**2x IRLZ44Z MOSFET N-Channel**

11:



**4x 10K Resistance**

12:



**2x 1K Resistance**

## **Software and Tools Used**

- Atmel Studio – Firmware development and Embedded C programming
- Proteus Design Suite – Circuit design and system Simulation
- Extreme Burner – Flashing firmware to the Microcontroller
- Zadig – USB driver installation for programming tools (AVR/USB interface)

## System Behavior Summary

- Secure authentication using password or physical key.
- Automatic LCD power saving system.
- Automatic return to main menu after inactivity.
- Hidden password input with toggle visibility feature.
- Multi-layer alarm triggering system.
- Full anti-theft and transportation protection Mode



### Legend

■ Entry / action

■ Output / state

■ Decision

■ Success

■ Alarm / error

■ Monitoring

■ Timeout / shortcut

■ Start / end

→ Normal flow

-----> Loop / return

-----> Alarm trigger

B key returns to main menu from any option at any time

Alarm can only be cleared using the physical key

Wrong PW ×3 · Vibration detected · Unauthorized door open → all trigger alarm

## System Flowchart

## Challenges Faced

- Stabilizing readings from vibration sensors (SW-520D).
- Designing reliable menu navigation system.
- Implementing accurate timing for auto-return and LCD timeout.
- Managing alarm logic without false triggers.
- Synchronizing multiple security inputs (sensors, keypad, key switch).
- Designing the system to continuously monitor security sensors in parallel with menu navigation and password handling without blocking critical alarm functionality.

## Future Improvements

- Add fingerprint authentication.
- Add GSM/Wi-Fi notification system.
- Store access logs in EEPROM or external memory.
- Add mobile application control.
- Integrate the safe with an external security/alarm system capable of automatically contacting authorities in case of theft attempts, forced opening, or unauthorized movement.
- Add a hidden camera module that captures images when an incorrect password is entered or when any security alarm is triggered.
- Add a GPS tracking system capable of transmitting the safe location to nearby smartphones through Bluetooth communication if the safe is stolen

